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- Kapitel 1: Übersicht über Unmanned Aerial Systems (UAVs) (1 W)
- Kapitel 2: Wesentliche Komponenten/Subsysteme eines UAV (1 W)
- Kapitel 3: Modellierung und Simulation von UAVs (1 W)
- Kapitel 4: Flight State Controller (PD Cascade) (1W)
- Kapitel 5: Pfadplanung (4 W)
- Kapitel 6: Perception und Sensing (1 W)
- **Kapitel 7: Navigation (2 W)**
- Kapitel 8: Multi-Vehicle Cooperation and Coordination (1 W)

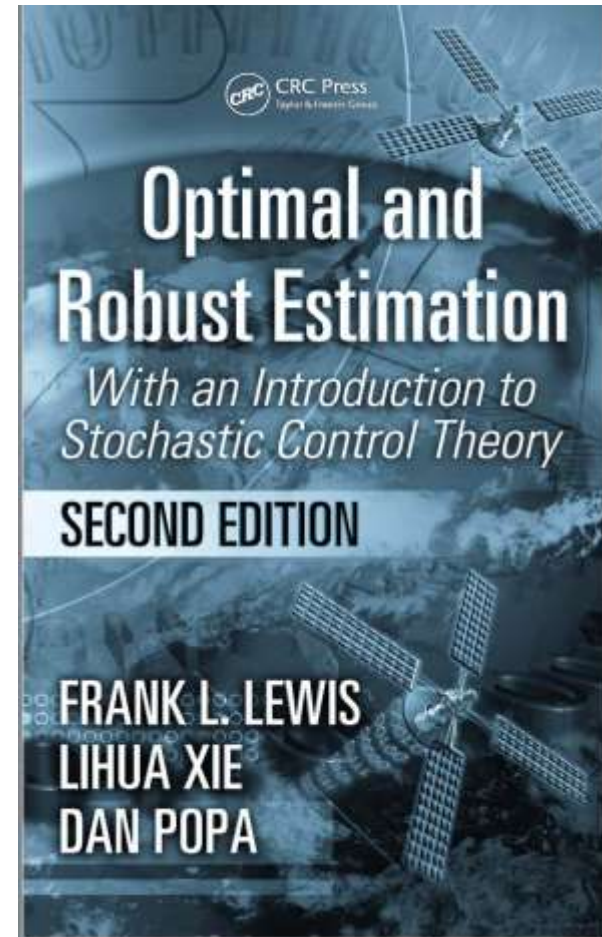


Frank L. Lewis (Life Fellow, IEEE) received the bachelor's degree in physics/electrical engineering and the M.S.E.E. degree from Rice University, Houston, TX, USA, in 1971, the M.S. degree in aeronautical engineering from the University of Florida, Gainesville, FL, USA, in 1977, and the Ph.D. degree from Georgia Tech, Atlanta, GA, USA, in 1988.

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Research Institute, University of Texas at Arlington, Arlington, TX, USA. He ranked at position 88 worldwide, 66 in the USA, and three in Texas of all scientists in computer science and electronics, by Guide2Research.com. He has 80 000 Google Scholar Citations. He works in feedback control, intelligent systems, reinforcement learning, cooperative control systems, and nonlinear systems. He is the author of seven U.S. patents, numerous journal special issues, 445 journal articles, 20 books, including the textbooks *Optimal Control*, *Aircraft Control*, *Optimal Estimation*, and *Robot Manipulator Control*.

Dr. Lewis received the Fulbright Research Award, the NSF Research Initiation Grant, the ASEE Terman Award, the International Neural Network Society Gabor Award, the U.K. Institute of Measurement and Control Honeywell Field Engineering Medal, the IEEE Computational Intelligence Society Neural Networks Pioneer Award, the AIAA Intelligent Systems Award, and the AACC Ragazzini Award. He is a Fellow of the National Academy of Inventors, IFAC, AAAS, the U.K. Institute of Measurement and Control, and PE Texas; and U.K. Chartered Engineer.



[Optimal and Robust Estimation - Google Books](#)

- Can also develop an **optimal estimator** for this type of system.
 - Given duality of regulator and estimator, would expect to see close connection between optimal estimator and regulator (LQR)
- Key step is to **balance** the effect of the various types of random noise in the system on the estimator:

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{B}_w\mathbf{w} \\ \mathbf{y} &= \mathbf{C}_y\mathbf{x} + \mathbf{v}\end{aligned}$$

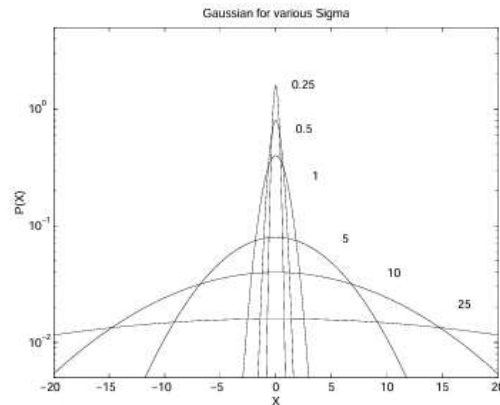
- $\mathbf{w}$: “process noise” – models uncertainty in the system model.
- $\mathbf{v}$: “sensor noise” – models uncertainty in the measurements.

- Typically assume that $\mathbf{w}(t)$ and $\mathbf{v}(t)$ are zero mean $E[\mathbf{w}(t)] = 0$ and
 - Uncorrelated Gaussian white random noises: no correlation between the noise at one time instant and another

$$E[\mathbf{w}(t_1)\mathbf{w}(t_2)^T] = R_{\mathbf{w}\mathbf{w}}(t_1)\delta(t_1 - t_2) \Rightarrow \mathbf{w}(t) \sim \mathcal{N}(0, R_{\mathbf{w}\mathbf{w}})$$

$$E[\mathbf{v}(t_1)\mathbf{v}(t_2)^T] = R_{\mathbf{v}\mathbf{v}}(t_1)\delta(t_1 - t_2) \Rightarrow \mathbf{v}(t) \sim \mathcal{N}(0, R_{\mathbf{v}\mathbf{v}})$$

$$E[\mathbf{w}(t_1)\mathbf{v}(t_2)^T] = 0$$



- With noise in the system, the model is of the form:

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} + B_w\mathbf{w}, \quad \mathbf{y} = C_y\mathbf{x} + \mathbf{v}$$

- And the estimator is of the form:

$$\dot{\hat{\mathbf{x}}} = A\hat{\mathbf{x}} + B\mathbf{u} + L(\mathbf{y} - \hat{\mathbf{y}}), \quad \hat{\mathbf{y}} = C_y\hat{\mathbf{x}}$$

- **Analysis:** in this case:

$$\begin{aligned}\dot{\tilde{\mathbf{x}}} &= \dot{\mathbf{x}} - \dot{\hat{\mathbf{x}}} = [A\mathbf{x} + B\mathbf{u} + B_w\mathbf{w}] - [A\hat{\mathbf{x}} + B\mathbf{u} + L(\mathbf{y} - \hat{\mathbf{y}})] \\ &= A(\mathbf{x} - \hat{\mathbf{x}}) - L(C_y\mathbf{x} - C_y\hat{\mathbf{x}}) + B_w\mathbf{w} - L\mathbf{v} \\ &= A\tilde{\mathbf{x}} - LC_y\tilde{\mathbf{x}} + B_w\mathbf{w} - L\mathbf{v} \\ &= (A - LC_y)\tilde{\mathbf{x}} + B_w\mathbf{w} - L\mathbf{v}\end{aligned}$$

- This equation of the estimation error explicitly shows the **conflict** in the estimator design process. Must **balance** between:
 - Speed of the estimator decay rate, which is governed by

$$\operatorname{Re}[\lambda_i(A - LC_y)]$$

- Impact of the sensing noise $\mathbf{v}$ through the gain L
- Fast state reconstruction requires rapid decay rate – typically requires a large L , but that tends to magnify the effect of $\mathbf{v}$ on the estimation process.
 - The effect of the process noise is always there, but the choice of L will tend to mitigate/accentuate the effect of $\mathbf{v}$ on $\tilde{\mathbf{x}}(t)$.

- Can predict the **mean square value** of the state (estimation error in this case) $Q(t) = E[\tilde{\mathbf{x}}(t)\tilde{\mathbf{x}}(t)^T]$ over time using $Q(0) = Q_0$ and

$$\begin{aligned}\dot{Q}(t) &= [A - LC_y] Q(t) + Q(t) [A - LC_y]^T \\ &\quad + \begin{bmatrix} B_w & -L \end{bmatrix} \begin{bmatrix} R_{ww} & 0 \\ 0 & R_{vv} \end{bmatrix} \begin{bmatrix} B_w^T \\ -L^T \end{bmatrix} \\ &= [A - LC_y] Q(t) + Q(t) [A - LC_y]^T + B_w R_{ww} B_w^T + L R_{vv} L^T\end{aligned}$$

- Instead, consider option of trying to minimize trace $\dot{Q}(t)$, the argument being that then $\int_0^t \text{trace } \dot{Q}(\tau) d\tau$ is small.
 - Not quite right, but good enough to develop some insights
- To proceed note that

$$\frac{\partial}{\partial X} \text{trace}[AXB] = \frac{\partial}{\partial X} \text{trace}[B^T X^T A^T] = A^T B^T$$

and

$$\frac{\partial}{\partial X} \text{trace}[AXBX^T C] = A^T C^T X B^T + C A X B$$

- So for minimum we require that

$$\frac{\partial}{\partial L} \text{trace } \dot{Q} = -2Q^T C_y^T + 2LR_{vv} = 0$$

which implies that

$$L = Q(t)C_y^T R_{vv}^{-1}$$

- Note that if we use this expression for L in the original differential Lyapunov Equation, we obtain:

$$\begin{aligned}
 \dot{Q}(t) &= [A - LC_y] Q(t) + Q(t) [A - LC_y]^T + B_w R_{ww} B_w^T + L R_{vv} L^T \\
 &= [A - Q(t) C_y^T R_{vv}^{-1} C_y] Q(t) + Q(t) [A - Q(t) C_y^T R_{vv}^{-1} C_y]^T \\
 &\quad + B_w R_{ww} B_w^T + Q(t) C_y^T R_{vv}^{-1} R_{vv} (Q(t) C_y^T R_{vv}^{-1})^T \\
 &= A Q(t) + Q(t) A^T - 2 Q(t) C_y^T R_{vv}^{-1} C_y Q(t) + B_w R_{ww} B_w^T \\
 &\quad + Q(t) C_y^T R_{vv}^{-1} C_y Q(t)
 \end{aligned}$$

$$\dot{Q}(t) = A Q(t) + Q(t) A^T + B_w R_{ww} B_w^T - Q(t) C_y^T R_{vv}^{-1} C_y Q(t)$$

which is obviously a matrix differential Riccati equation.

- **Goal:** develop an estimator $\hat{\mathbf{x}}(t)$ which is a linear function of the measurements $\mathbf{y}(\tau)$ ($0 \leq \tau \leq t$) and **minimizes** the function

$$J = \text{trace}(Q(t))$$
$$Q(t) = E [\{\mathbf{x}(t) - \hat{\mathbf{x}}(t)\} \{\mathbf{x}(t) - \hat{\mathbf{x}}(t)\}^T]$$

which is the **covariance** for the **estimation error**.

$$\dot{\hat{\mathbf{x}}}(t) = A\hat{\mathbf{x}} + L(t)(\mathbf{y}(t) - C_y\hat{\mathbf{x}}(t))$$

where $L(t) = Q(t)C_y^T R_{vv}^{-1}$ and $Q(t) \geq 0$ solves

$$\dot{Q}(t) = AQ(t) + Q(t)A^T + B_w R_{ww} B_w^T - Q(t)C_y^T R_{vv}^{-1} C_y Q(t)$$

- Note that $\hat{x}(0)$ and $Q(0)$ are known
- Differential equation for $Q(t)$ **solved forward in time**.
- **Filter form** of the **differential matrix Riccati** equation for the **error covariance**.
- Note that the $AQ(t) + Q(t)A^T \dots$ is different than with the regulator which had $P(t)A + A^T P(t) \dots$

- Note that an increase in $Q(t)$ corresponds to **increased uncertainty in the state estimate**. $\dot{Q}(t)$ has several contributions:
 - $AQ(t) + Q(t)A^T$ is the homogeneous part
 - $B_w R_{ww} B_w^T$ increase due to the process measurements
 - $Q(t)C_y^T R_{vv}^{-1} C_y Q(t)$ decrease due to measurements
- The estimator gain is $L(t) = Q(t)C_y^T R_{vv}^{-1}$
 - Feedback on the **innovation**, $y - \hat{y}$
 - If the uncertainty about the state is high, then $Q(t)$ is large, and so the innovation $y - C_y \hat{x}$ is weighted heavily ($L \uparrow$)
 - If the measurements are very accurate $R_{vv} \downarrow$, then the measurements are heavily weighted

1. $R_{vv} > 0, R_{ww} > 0$
 2. All plant dynamics are constant in time
 3. $[A, C_y]$ detectable
 4. $[A, B_w]$ stabilizable
- Then, as with the LQR problem, the covariance of the LQE quickly settles down to a constant Q_{ss} independent of $Q(0)$, as $t \rightarrow \infty$ where

$$AQ_{ss} + Q_{ss}A^T + B_w R_{ww} B_w^T - Q_{ss} C_y^T R_{vv}^{-1} C_y Q_{ss} = 0$$

- Stabilizable/detectable gives a unique $Q_{ss} \geq 0$
 - $Q_{ss} > 0$ iff $[A, B_w]$ controllable
 - $L_{ss} = Q_{ss} C_y^T R_{vv}^{-1}$
- If Q_{ss} exists, the steady state filter

$$\begin{aligned}\dot{\hat{\mathbf{x}}}(t) &= A\hat{\mathbf{x}} + L_{ss}(\mathbf{y}(t) - C_y\hat{\mathbf{x}}(t)) \\ &= (A - L_{ss}C_y)\hat{\mathbf{x}}(t) + L_{ss}\mathbf{y}(t)\end{aligned}$$

is asymptotically stable iff (1)–(4) above hold.

- Given that $\dot{\hat{\mathbf{x}}} = (A - LC_y)\hat{\mathbf{x}} + L\mathbf{y}$
- Consider a scalar system, and take the Laplace transform of both sides to get:

$$\frac{\hat{X}(s)}{Y(s)} = \frac{L}{sI - (A - LC_y)}$$

- This is the transfer function from the “measurement” to the “estimated state”
 - It looks like a low-pass filter.
- Clearly, by lowering R_{vv} , and thus increasing L , we are pushing out the pole.
 - DC gain asymptotes to $1/C_y$ as $L \rightarrow \infty$

